```
import socket
import struct
# Create RAW Socket
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROT
# Local machine IP
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))
# Include IP headers
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
# Enable promiscuous mode (Windows)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

print("Packet Sniffer Started...\n")
try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)
        ip_header = raw_data[0:20]
        iph = struct.unpack("!BBHHHBBH4s4s", ip_header)

        version = iph[0] >> 4
        ihl = iph[0] & 0x0F
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])

        print("----------------------------------------")
        print("Version       :", version)
        print("Header Length:", ihl * 4, "bytes")
        print("TTL           :", ttl)
        print("Protocol      :", protocol)
        print("Source IP     :", source)
        print("Destination   :", destination)
        print("Packet Length:", len(raw_data))

except KeyboardInterrupt:
    # Disable promiscuous mode before exiting
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

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Exp No: 05

Date: 01/08/26

Develop a customized ping command to test the server connectivity.

Aim: To develop a customized ping command in Python that checks the connectivity between a client and server by sending ICMP Echo Requested Packets and displaying the responce time and status.

Procedure:-

Step 1:- Set up environment & confirm Python 3 in installed.

Step 2: Understand Packets structure you will Parse

Step 3:- Create raw socket, create a file sniffer Py & Start.

Start 4: Parse ethernet header, add tune to unpack first 14 bytes.

Step 5:- Parse the IP header and

Step 6:- Parse TCP / UDP / icmp header

Step 7:- write main sniffing Loop

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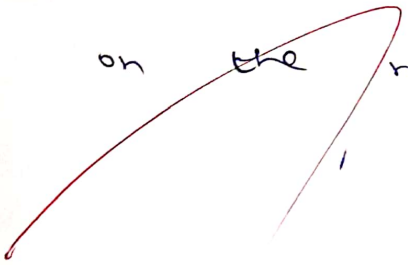
```
Version       : 4
Header Length: 20 bytes
TTL           : 62
Protocol      : 17
Source IP     : 142.251.222.206
Destination   : 172.16.10.48
Packet Length: 150
-----------------------------------
Version       : 4
Header Length: 20 bytes
TTL           : 62
Protocol      : 17
Source IP     : 142.251.222.206
Destination   : 172.16.10.48
Packet Length: 53
-----------------------------------
Version       : 4
Header Length: 20 bytes
TTL           : 62
Protocol      : 17
Source IP     : 142.251.222.206
Destination   : 172.16.10.48
Packet Length: 53
-----------------------------------
Version       : 4
Header Length: 20 bytes
TTL           : 128
Protocol      : 17
Source IP     : 172.16.10.48
Destination   : 142.251.222.206
Packet Length: 63
-----------------------------------
Version       : 4
Header Length: 20 bytes
TTL           : 62
Protocol      : 17
Stopped
```

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Step 8:- Run the sniffer.

Step 9:- Stop the sniffer with Ctrl + c

Step 10:- Display reable or unreable cable based on the result.



Result:-

The network connectivity to the specified host was successfully tested, and the server status was displayed correctly.

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